

GRAPHIC ERA HILL UNIVERSITY, BHIMTAL

OOPS ASSIGNMENT : 1

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Aim

To design and develop a functional **calculator application using C++ and the Qt framework**, providing a graphical user interface for performing basic arithmetic operations.

Problem Statement

Develop a **GUI-based calculator** that allows users to perform basic mathematical operations such as **addition, subtraction, multiplication, and division**. The application should provide an easy-to-use interface with buttons for numerical input, operators, equal, clear, and other basic controls.

TECHNOLOGIES USED

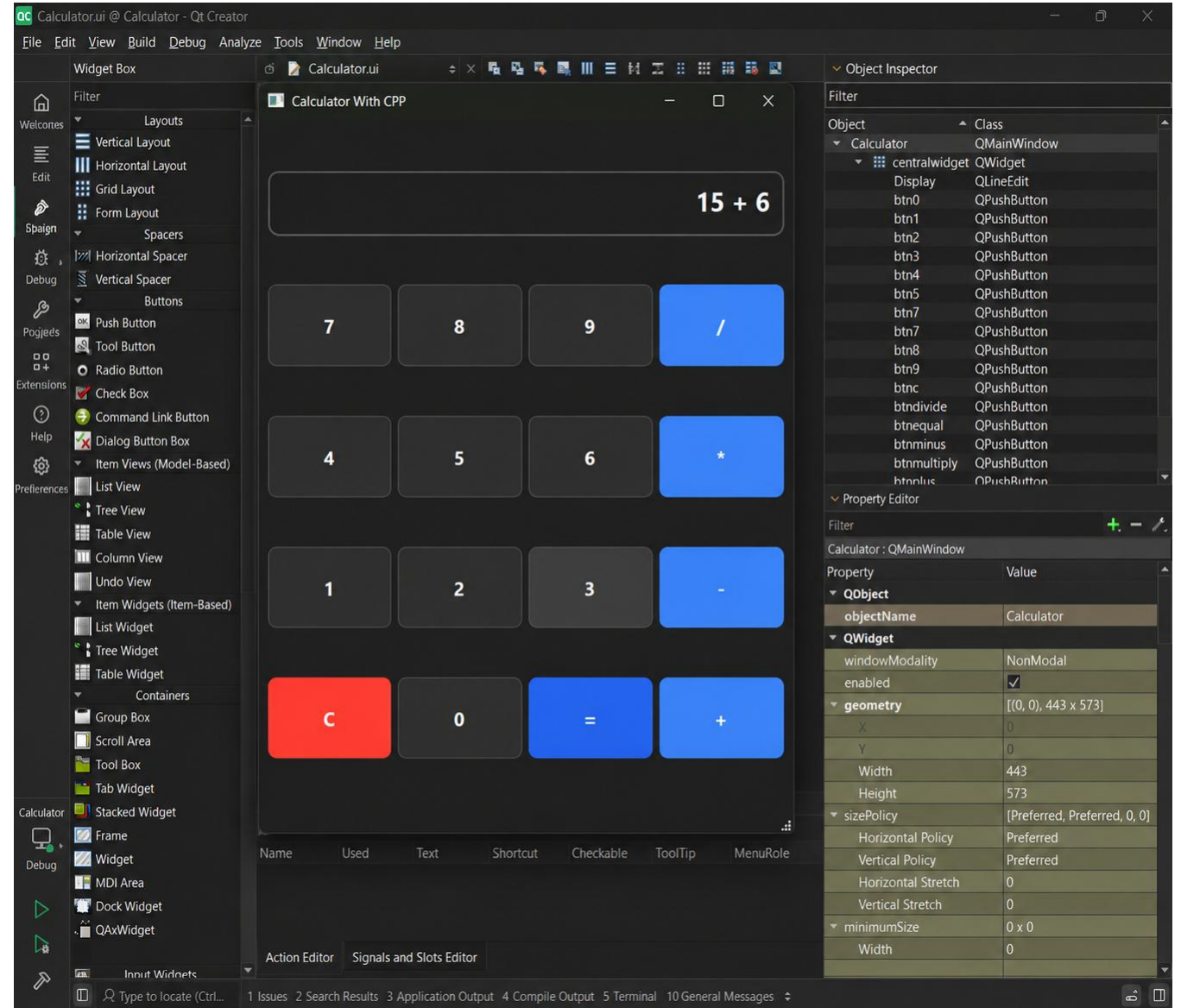
S. No.	Technology	Purpose
1.	C++	Core programming and calculator logic
2.	Qt 6	GUI development
3.	Qt Designer	Designing the calculator interface
4.	Qt Creator	Development and execution
5.	CMake	Project build and configuration
6.	Git & GitHub	Version control and project hosting

EXPLANATION

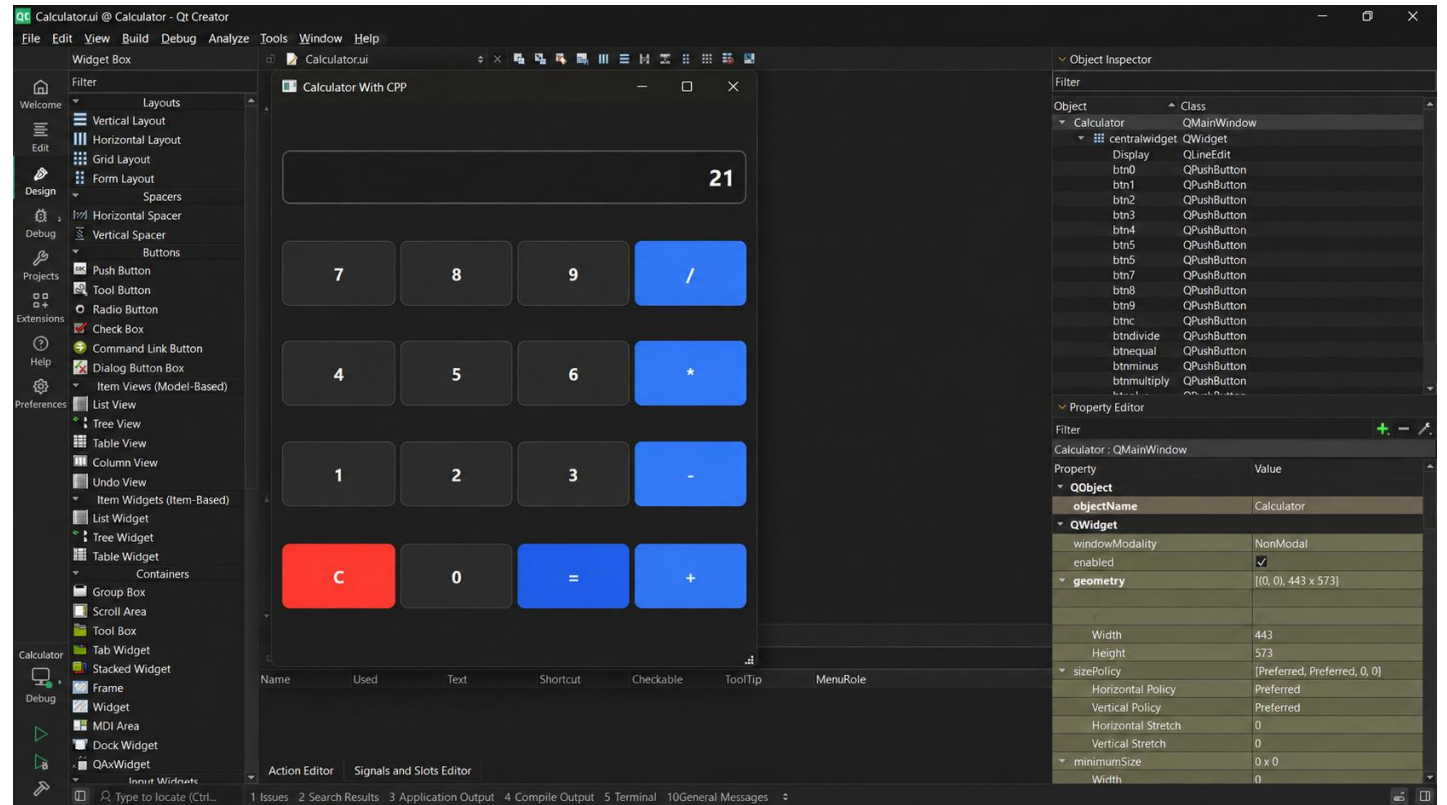
The **Calculator** is a GUI-based desktop application developed using **C++ and the Qt framework**. It provides a simple and user-friendly interface with buttons for numbers, arithmetic operators, decimal values, equal, clear, and backspace. The application performs basic operations such as **addition, subtraction, multiplication, and division**. Qt's **signal-slot mechanism** is used to handle button clicks and connect them with the appropriate functions. The calculator also handles errors such as **division by zero** and displays the result on the screen

SOURCE CODE: [**GITHUB**](#)

SCREENSHOT 1



SCREENSHOT 2



Q : WAP TO CHECK PALINDROME OR NOT?

```
#include <iostream>
using namespace std;

int main()
{
    int n, original, reversed = 0, digit;

    cout << "Enter a number: ";
    cin >> n;

    original = n;

    while (n != 0)
    {
        digit = n % 10;
        reversed = reversed * 10 + digit;
        n = n / 10;
    }

    if (original == reversed)
        cout << "Palindrome number";
    else
        cout << "Not a palindrome number";

    return 0;
}
```

Q : WAP TO CHECK FACTORIAL OF A GIVEN NUMBER

```
#include <iostream>
using namespace std;

int main()
{
    int n;
    long long factorial = 1;

    cout << "Enter a number: ";
    cin >> n;

    for (int i = 1; i <= n; i++)
    {
        factorial *= i;
    }

    cout << "Factorial = " << factorial;

    return 0;
}
```